FOUNDATIONS + WATERSHEDS [M1-M2]

Hydrology: study of water occurrence, movement, distribution, and properties. Water occurs as solid, liquid, and vapour in atmosphere, surface water, groundwater, and cryosphere.

Scientific method: observation/question, testable hypothesis, prediction, controlled evidence, analysis, conclusion, revision. Conclusions remain provisional.

System: defined boundary plus time interval. **Input:** enters. **Output:** leaves. **Storage:** remains within the boundary. Classification changes if the boundary changes.

Steady state: no net storage change, so total inputs equal total outputs. Flows can remain nonzero.

Storage vs. flux: lake level, soil water, groundwater, snow, and glacier ice are storages. P, ET, runoff, recharge, and discharge are fluxes.

Watershed/catchment/drainage basin: topographically bounded area draining to a common outlet. **Divide:** boundary. **Tributary:** smaller channel entering a larger one.

Hydrologic controls: solar energy drives phase changes and ET; gravity and pressure/elevation gradients drive water movement.

Global water: oceans hold about 97%; freshwater is under 3%; most freshwater is frozen or underground. Glaciers and ice caps are the largest freshwater reserve.

ATMOSPHERIC WATER + PRECIPITATION [M3]

Evaporation: liquid to vapour. **Condensation:** vapour to liquid. **Sublimation:** solid to vapour. **Deposition:** vapour to solid.

Saturation: actual vapour pressure equals saturation vapour pressure. At saturation RH = 100% and VPD = 0.

Dew point: temperature at which cooling air becomes saturated. Cooling lowers e_s; if actual e stays fixed, RH rises.

Adiabatic cooling: rising air expands and cools without exchanging heat. Condensation also requires nuclei such as dust or salt.

Convective precipitation: surface heating creates intense, local uplift. **Orographic:** terrain uplift produces wet windward and dry rain-shadow sides.

Frontal precipitation: warm and cold air meet. Cold fronts are steeper, shorter, and more intense; warm fronts are gentler, longer, and less intense.

Convergence: colliding air masses are forced upward. Frontal lifting dominates southern Ontario's middle latitudes.

Global circulation: warm rising equatorial air favours rain; descending warming air near 30° N/S creates arid belts. Deserts may be hot or polar.

Rain vs. snow: liquid enters active flow quickly; snow stores water until melt. Wind can strongly redistribute snow after it falls.

Snow water equivalent: liquid-water depth contained in snow. Fresh snow is low density; wind-packed snow is denser.

PRECIPITATION MEASUREMENT [M4]

Non-recording gauge: manual accumulated depth with no intensity history. **Recording gauge:** weighing, float, or tipping-bucket system records timing.

Tipping bucket: gives event timing and intensity, but very small droplets may not tip the bucket. Mechanical or electronic failure creates gaps.

Gauge errors: wind under-catch, evaporation, splashing, poor placement, observer error, station movement, and equipment changes.

Gauge siting: flat, open, low-wind location; obstacles should be at least four times their height away. Snow gauges are elevated and shielded.

Arithmetic mean: first exclude every gauge outside the study-area boundary, then give each inside gauge equal weight. Best for flat, uniform terrain. **Thiessen:** station-area weights. **Isohyetal:** contour-band weights.

Dundee's repeated M4 trap: a gauge shown on the map is not automatically included. Example: inside values 0.8, 1.6, 1.2 cm give (0.8 + 1.6 + 1.2)/3 = 1.2 cm; ignore outside values 0.9 and 0.7 cm.

Interpolation: estimates values between observations. It extends spatial coverage but remains modelled, not directly measured, data.

M1-M4 QUICK DISTINCTIONS

Accuracy: closeness to the accepted/true value. **Precision:** repeatability or resolution. Extra decimal places do not remove systematic error.

Representativeness: whether a point measurement describes the larger area. More well-sited gauges can improve spatial accuracy more than one extremely precise gauge.

Quality control: calibration, consistent siting, metadata, maintenance, overlap during equipment changes, and comparison with nearby stations.

Recording vs. non-recording: recording devices preserve intensity and timing; non-recording gauges can still provide reliable totals with correct observation.

Rainfall variability: convective storms are localized; frontal storms are broader; orographic effects follow terrain; snowfall is readily redistributed by wind.

Arid-region threshold: the course uses no more than 25 cm annual precipitation. Low precipitation does not require high temperature.

EVAPORATION + TRANSPIRATION [M5]

Evapotranspiration: evaporation from free water, interception, soil, and other wet surfaces plus plant transpiration. Condensation of dew or frost is opposite to ET.

Primary control: available energy. Warm, dry, sunny, windy conditions favour E/ET; energy limits the maximum.

Vapour-pressure deficit: atmospheric drying demand. Positive VPD favours evaporation; high RH lowers VPD and evaporation.

Wind: removes humid boundary air. **Albedo:** higher albedo reflects more radiation and tends to reduce evaporation.

Water-body controls: fresh water evaporates faster than saline water. Shallow water heats quickly; deep water stores heat and can sustain cool-season or nighttime evaporation.

Lake fetch: moving dry air humidifies over a long lake, so average evaporation rate may decline downwind even while total lake loss is large.

Plant types: hydrophyte roots in water; mesophyte uses ordinary soil moisture; xerophyte is arid-adapted with shallow dense roots; phreatophyte reaches the water table.

Seasonality: transpiration peaks in summer and approaches zero in winter; Canadian total ET normally peaks in summer.

MEASURING EVAPOTRANSPIRATION [M6]

Potential ET: maximum with unlimited water and unsaturated air. **Actual ET:** ET under existing water and atmospheric conditions; actual ET cannot exceed potential ET.

Evaporation pan: estimates potential open-water evaporation; correct for rainfall, added/removed water, and pan coefficient.

Lysimeter: isolated soil/vegetation sample. Weighing type uses mass loss; water-balance type tracks all flows and storage.

Chamber: measures local vapour change. Vegetated chamber loss minus comparable covered bare-surface loss estimates transpiration.

Energy balance: partitions incoming radiation among reflection, emitted longwave, latent heat, sensible heat, and plant growth.

Empirical model: combines calibrated coefficient, wind, and vapour-pressure deficit; local calibration may not transfer to unlike sites.

Groundwater fluctuation method: uses nighttime water-table recovery, signed daily change, and specific yield; requires shallow plant-accessible groundwater.

WATER BALANCE CONCEPTS [M7]

Positive balance: storage increases. **Negative balance:** storage decreases. Sign interpretation is part of every answer.

Entire watershed: precipitation and groundwater inflow may enter; ET, stream discharge, and groundwater outflow may leave. Surface-water inflow is not mandatory when the watershed boundary follows divides.

Lake balance: convert P and E depths over lake area into volumes before combining them with surface-water and groundwater volume rates.

Accuracy rule: match system boundary, area, time interval, and units before substitution. Round at the end.

INSTRUMENT + GRAPH RECOGNITION

Hydrograph: discharge versus time and may overlay precipitation. **Hyetograph:** precipitation intensity versus time.

Stilling well: stabilizes water level for stage measurement, commonly with float and shaft encoder. **Piezometer:** measures hydraulic head at a location/depth.

Rating curve: converts locally measured stage to discharge. It is valid near the calibrated section and must be updated if channel geometry changes.

M5-M7 QUICK ANSWERS

Evaporation requirement: energy plus positive vapour-pressure deficit in the overlying air. Saturation or high RH suppresses net evaporation.

Strongest E/ET factor: energy from the sun. Air temperature, humidity, and wind modify the response.

Large vs. small lake: dry air becomes more humid across a long fetch, so a larger lake does not automatically have a higher average evaporation rate.

Summer lake maximum: large, shallow, freshwater combines rapid warming, extensive exposed area, and no salinity suppression.

ET inclusion: free-water, soil, and interception evaporation; plant transpiration; sublimation where applicable. Condensation is excluded.

Field evaporation: standard answer is an evaporation pan. A rain gauge measures precipitation, not evaporation.

Soil control: rapid drainage can move water below the evaporation zone; high permeability alone does not guarantee greater long-term evaporation.

Groundwater diurnal cycle: daytime ET commonly lowers the water table; nighttime recovery occurs when ET weakens. A daytime rise is the wrong direction.

Steady-state simplification: remove ΔS, then omit only the fluxes explicitly stated as negligible or absent.

CALCULATION CORE [M3-M12]

Water balance, volume, and units

ΔS = inputs - outputs

$\Delta S = (P + Q_{in} + G_{in}) - (ET + Q_{out} + G_{out})$

Steady state: $\Delta S = 0$ and Σ inputs = Σ outputs

Volume $V = dA$; rate-volume $V = Qt$; level change $\Delta h = \Delta S/A$

1 mm over 1 km^2 = 1000 m^3; 1 day = 86,400 s; 1 m^3 = 1000 L

Lake example: 5 km^2 × (12 - 8) mm/day = 20,000 m^3/day; net other flows +9 gives ΔS = +20,009 m^3/day

Level example: 31,000 m^3 / 2,000,000 m^2 = 0.0155 m = 15.5 ≈ 16 mm

Air, lapse rates, and snow

$RH = (e/e_s) \times 100\%$; $VPD = e_s - e_a$

$\Delta T = -\Gamma \Delta z$; dry Γ = 1.0°C/100 m; wet = 0.6°C/100 m; environmental = 0.65°C/100 m

D_s = SWE/snow depth; SLR = snow depth/SWE = $1/D_s$; SWE = depth/SLR

Precipitation data

Intensity $I = P/\Delta t$

Missing station $P_x = (N_x/n)\Sigma(P_i/N_i)$

Arithmetic $\bar{P} = (1/G_{in})\Sigma P_{inside}$; exclude outside gauges. Thiessen $\bar{P} = \Sigma w_i P_i$; $\Sigma w_i = 1$

Isohyetal volume $V = \Sigma(A_{band} \bar{P}_{band})$; depth = total V / total A

ET and soils

Pan E = corrected depth loss × K_p / time

ET water balance: $E = P + SW_{in} + GW_{in} - SW_{out} - GW_{out} - \Delta S$

Groundwater $ET = S_y(24a + b)$; keep b signed

$V_t = V_s + V_v$; $V_v = V_w + V_a$; $m_t = m_s + m_w$

Porosity $n = V_v/V_t$; void ratio $e = V_v/V_s$; saturation $S_r = V_w/V_v$

Water content $w = m_w/m_s$; bulk density $\rho = m_t/V_t$; PI = LL - PL

Groundwater and rivers

Water-table elevation $z_{wt} = z_{ground} - d_{water}$; head $h = z + h_p$

Pressure $p = \rho g h_p$; gradient $i = \Delta h/L$

Darcy $q = Ki$; discharge $Q = qA = KiA$; pore velocity $v = q/n$; travel time $t = L/|v|$

Subsection $A_i = w_i d_i$; mean $\bar{v}_i = (v_{0.2D} + v_{0.8D})/2$

$Q_i = A_i \bar{v}_i$; $Q_{total} = \Sigma Q_i$; 1 m^2 × 1 m/s = 1 m^3/s

Stage $H_{stage} = H - H_0$; return probability P = 1/T or 100/T percent

Climate

Constant-rate sea-level baseline $\Delta h = rt$; convert mm to cm or m after multiplying

VARIABLE KEY

Balance: P precipitation; ET evapotranspiration; Q surface flow/discharge; G groundwater flow; ΔS storage change; A area; V volume; d or Δh depth/change; t time.

Air: e actual vapour pressure; e_s saturation vapour pressure; RH relative humidity; VPD deficit; Γ lapse rate; Δz elevation change; SWE snow-water equivalent; SLR snow-liquid ratio.

Ground/river: h hydraulic head; z elevation head; h_p pressure head; K conductivity; i gradient; q specific discharge; n porosity; v pore velocity; T return period.

CALCULATION + GRAPH TRAPS

Depth and volume: never add mm/day directly to m^3/day. Convert the depth over the stated area first.

Rate and total: discharge is m^3/s; multiply by elapsed seconds for volume. Keep monthly day counts explicit.

Signs: inputs positive, outputs negative. A negative ΔS is a storage decrease. Preserve the sign of groundwater-ET term b.

Darcy: q is not pore velocity. Divide by decimal porosity to obtain v; use absolute velocity magnitude in travel time.

Hydrostatic pressure: use vertical water-column height, not total well depth unless the water column fills the well.

Vapour graph: read e_s at the given temperature; compare measured e with e_s before deciding evaporation, equilibrium, or condensation.

Rating graph: move from stage axis to the rating curve and then to discharge. Do not treat stage as discharge.

Return period: probability repeats each year. Independent years do not schedule the event.

SOILS + UNSATURATED ZONE [M8]

Soil: mineral matter, organic matter, water, and air. **Regolith:** soil and loose rock above bedrock.

Formation controls: parent material, time, climate, organisms, and topography. Soil is a dynamic hydrologic storage and filter.

Profile: O organic; A mineral topsoil; E eluviation; B accumulation/subsoil; C weathered parent; R bedrock. O + A are topsoil.

Weathering: breaks material down. **Erosion:** removes and transports it. Mechanical weathering keeps composition; chemical weathering changes minerals.

Porosity: fraction of total volume that is void space. **Permeability:** ability of connected pores to transmit fluid. Clay can have high porosity and low permeability.

Void ratio: void volume divided by solid volume. **Saturation:** water volume divided by void volume. **Water content:** water mass divided by dry-solid mass.

Specific yield: drainable water. **Specific retention:** water held against gravity. Course relationship: yield + retention = porosity.

Consistency limits: shrinkage, plastic, and liquid limits separate solid, semi-solid, plastic, and liquid behaviour. PI = LL - PL.

Infiltration: entry into soil. **Percolation:** downward subsurface movement toward the saturated zone. **Recharge:** addition to groundwater.

Vadose zone: above water table and unsaturated. **Saturated zone:** pores filled with water. The course figure groups the capillary fringe with saturated materials.

Degradation: erosion, desertification, contamination, topsoil loss, fertilizer overuse, and intensified agriculture. Controls include windbreaks, terraces, and contouring.

GROUNDWATER SYSTEMS [M9]

Water table: top of the saturated zone. It can rise after recharge and fall during pumping, drainage, and ET.

Aquifer: stores and transmits water. **Aquitard/confining layer:** restricts flow. Types include unconfined, confined, semi-confined/leaky, and karstic.

Potentiometric surface: level to which confined water rises. If above land surface, the well flows artesian.

Hydraulic head: elevation head plus pressure head. Groundwater flows from high head to low head, not necessarily from high ground to low ground.

Darcy specific discharge q: bulk flow per total area. **Pore velocity v:** faster actual water speed through pore space, $v = q/n$.

Gaining stream: receives groundwater. **Losing stream:** supplies groundwater.
Flow-through: groundwater enters one bank and exits another.

Perched losing stream: separated from the regional water table by unsaturated material. One local well cannot map an entire aquifer.

Contaminant sources: landfills, septic systems, farms/manure, road salt, spills, pesticides, pharmaceuticals, and saline intrusion.

Walkerton: manure-derived contamination entered open Well 5 through fractured karst-connected aquifers.

M8-M9 QUICK DISTINCTIONS

Sand vs. clay: sand usually has larger connected pores, faster drainage, and lower retention; clay can retain more water despite low permeability.

Hydraulic gradient vs. conductivity: i describes head change per distance; K describes material transmission ability. Darcy flow depends on both.

Unconfined aquifer: upper boundary is the water table. **Confined aquifer:** bounded by low-permeability layers and commonly pressurized.

Karst/fractures: rapid preferential flow reduces natural filtration and can move contaminants quickly.

Groundwater accuracy: infer conditions from wells, head, geology, K, tracers, and models. Interpolation between wells adds uncertainty.

Contaminant analysis: identify source, pathway, velocity, storage, receptor, and whether pumping changes the gradient.

RUNOFF + RIVER SYSTEMS [M10]

Runoff components: direct precipitation, shallow interflow, groundwater discharge/baseflow, and snowmelt.

Runoff controls: climate, catchment size, precipitation type/intensity/distribution, hydraulic structures, vegetation, and urbanization.

Urbanization: impervious cover reduces infiltration, speeds routing, increases peak discharge and flood risk, steepens the rising limb, and shortens lag time.

Stage: water-surface height above a datum. **Discharge:** water volume passing a cross-section per unit time. **Wetted perimeter:** channel boundary touching water.

Velocity distribution: slowest near rough bed/banks; fastest near the centre just below the surface, not at the surface.

Velocity-area method: divide the channel into subsections and sum area times representative velocity. One reading at 0.6D is acceptable but less accurate than 0.2D and 0.8D.

Meander: curved channel path. Outer bank has faster flow and erosion; inner bank has slower flow and deposition. A cutoff can form an oxbow lake on the floodplain.

Competence: largest particle transported. **Capacity:** total sediment load. Faster water generally increases both.

Sediment modes: bed load by traction and saltation; fine grains in suspended load; ions in dissolved load. Turbulent load is not a formal transport mode.

Hydrograph: discharge response through time. Baseflow persists between storms; stormflow/quickflow is the event response above baseflow.

Return period: inverse annual probability. A 100-year event has a 1% chance each year, not one guaranteed occurrence per century.

Gradient: commonly steepest near headwaters/source. Natural streamflow is mainly turbulent.

GLACIERS + GLACIATION [M11]

Glacier: persistent compacted/recrystallized snow that flows under its own weight. Alpine glaciers are topographically confined; continental ice sheets cover vast land.

Positive budget: accumulation exceeds ablation. **Negative budget:** ablation exceeds accumulation. Equilibrium line separates the zones.

Movement: internal plastic deformation is the main mechanism; basal sliding is aided by meltwater. Speed depends on slope, ice volume, and temperature.

Terminus: glacier toe. Ice may keep flowing downslope while the terminus retreats upslope when ablation exceeds supply.

Milankovitch cycles: eccentricity changes orbit shape; obliquity changes axial tilt; precession changes axis direction. They alter insolation and help explain glacial cycles.

Alpine erosion: U-shaped valley, hanging valley, cirque, arete, and horn. Eskers are depositional, not erosional alpine features.

Moraines: lateral at sides; medial where lateral moraines join; terminal/end at maximum toe; recessional marks pauses during retreat.

Continental deposits: drumlin streamlined in ice-flow direction; esker sinuous sand/gravel ridge; kame conical hill; kettle depression; outwash meltwater sediment.

Isostatic depression: crust sinks under ice load. **Isostatic rebound:** slow uplift after melting. Last glacial maximum was about 20,000 years ago.

M10-M11 QUICK ANSWERS

River level term: stage. Discharge requires average velocity and cross-sectional area.

Stilling-well purpose: steady water-elevation reading, typically using a float and shaft encoder.

Hydrograph capability: may display both precipitation and river discharge through time.

Runoff-factor question: climate, catchment size, precipitation characteristics, structures, vegetation, and urbanization all matter.

Past ice direction: elongated drumlins, eskers, moraines, striations, and related depositional patterns can indicate movement.

Valley-glacier speed: chiefly related to slope, ice volume/thickness, and temperature rather than sediment load or terminus meltwater alone.

Glacial budget: compare water-equivalent accumulation with ablation, not the fraction of rock or till in the ice.

CLIMATE CHANGE + HYDROLOGY [M12]

Climate: long-term pattern measured over at least 30 years. One storm, season, or year is weather.

Greenhouse effect: atmosphere transmits incoming short-wave radiation more readily and absorbs outgoing long-wave radiation, warming air, land, and ocean.

Important greenhouse gas: CO_2; the course also names CH_4, nitrous oxide, and CFCs. Carbon monoxide and sulphur dioxide are not the quiz answer.

Natural drivers: solar output, orbit/tilt, plate tectonics, and volcanoes. Major/larger hurricanes are effects, not climate drivers.

Proxy records: ice cores, corals, tree rings, and sediment layers. Ice cores and direct observations show current CO_2 higher than any point in 800,000 years.

Course values: about 280 ppm preindustrial; just under 420 ppm in July 2022; human annual CO_2 about 60 times volcanic emissions.

Quiz 12 anchors: doubled current CO_2 gives about +3°C global warming, plus or minus 1°C. Retain the saved quiz's 1867/Svante Arrhenius pairing.

Documented hydrologic changes: warmer surface waters, reduced ice cover, earlier snowmelt, increased high-latitude runoff, heavier precipitation events, and shorter ice seasons.

Diurnal range: decreased from 1950-2004 and then changed little from 1979-2004 because maximum and minimum temperatures increased at similar rates.

IPCC: Intergovernmental Panel on Climate Change, established by the UN in 1988 to synthesize climate research and project future change.

Mitigation: reduces causes/emissions. **Adaptation:** manages impacts. Natural-plus-human models match observed warming better than natural-only models.

Sea level: almost 4 mm/year and accelerating, versus about 1-2 mm/year early in the twentieth century. Rising seas threaten coastal aquifers through saltwater intrusion.

HIGH-FREQUENCY TRUE/FALSE TRAPS

TRUE: steady state can have nonzero flows; a hydrograph can show P and Q; oxbow lakes occur on floodplains; glaciers are the largest freshwater reserve.

FALSE: every watershed requires surface-water inflow; condensation is ET; higher albedo increases evaporation; high humidity increases evaporation.

FALSE: permeability and porosity are identical; groundwater follows ground elevation alone; Darcy q equals pore velocity when n is below 1.

FALSE: river capacity is largest grain; maximum velocity is at the surface; turbulent load is a sediment mode; return period is a schedule.

FALSE: isostatic rebound is sinking; alpine glaciers carve V-shaped valleys; kames are erosional; eskers are alpine erosional features.

FALSE: major hurricanes drive climate; diurnal range is increasing at 0.07°C/decade; a short warm period proves climate change.

EXAM DECISION CUES

Definition question: identify the property requested. Largest grain = competence; total load = capacity; water height = stage; volume/time = discharge.

Calculation question: write the governing equation, convert all terms to one unit/time basis, preserve signs, substitute, interpret gain/loss, then round.

Graph question: read axes and units first. Saturation curve gives e_s; rating curve maps stage to Q; hydrograph plots Q with time; hyetograph plots P intensity.

Method question: arithmetic = equal weights after excluding outside gauges; Thiessen = gauge areas; isohyetal = spatial contours; 0.2D/0.8D = velocity-area discharge.

Instrument question: pan = potential open-water E; lysimeter = sample ET; stilling well = stable stage; piezometer = head; tipping bucket = rainfall timing/intensity.

OFFICIAL QUIZ ANCHORS

M5: evaporation requires energy and VPD; saturation vapour pressure above measured vapour pressure permits evaporation; increasing albedo decreases evaporation.

M5: xerophyte = arid climate with shallow dense roots; evaporation pan = field device; condensation of dew/frost is not ET.

M7: steady state means no overall storage change; for a complete watershed, surface-water inflow is not always required.

M8: permeability describes transmission, porosity describes void space, and high permeability does not automatically mean more evaporation.

M9: hydraulic head controls groundwater direction; hydrostatic pressure uses $\rho g h$; total head includes elevation and pressure.

M10: stage = river-water height; competence = maximum grain; capacity = total load; meanders are curved channel paths.

M10: oxbow lakes occur within floodplains; peak velocity is just below the surface; turbulent load is not a sediment mode.

M11: kames are conical depositional hills; lateral moraines lie along glacier sides; eskers are depositional ridges.

M11: Milankovitch cycles explain glacial-period entry/exit; internal plastic deformation drives most glacier movement; rebound is uplift.

M12: CO_2 is the greenhouse-gas answer; climate minimum is over 30 years; proxy sources include ice, coral, trees, and sediments.